Ageing Research Reviews 87 (2023) 101929

Contents lists available at ScienceDirect

Ageing Research Reviews

journal homepage: www.elsevier.com/locate/arr

Review

Fungi as a source of bioactive molecules for the development of longevity medicines

Yuliia Shevchuk [a], Kim Kuypers [b], Georges E. Janssens [a,c,*]

[a] Amsterdam UMC Location University of Amsterdam, Laboratory Genetic Metabolic Diseases, Meibergdreef 9, Amsterdam, the Netherlands
[b] Department of Neuropsychology and Psychopharmacology, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, the Netherlands
[c] Amsterdam Gastroenterology, Endocrinology and Metabolism, Amsterdam, the Netherlands

ARTICLE INFO

Keywords:
Medicinal mushroom
Medicinal fungi
Aging
Longevity medicine
Geroprotectors

ABSTRACT

Age-related loss of function brings age-related disease, and therefore it is of primary importance to search for interventions that can help minimize detrimental symptoms people deal with in old age. Fungi have always been given a great deal of attention and reverence in traditional medical practices for their ability to minimize harmful symptoms of diseases. More recently, the role of fungi in influencing healthspan and lifespan is being directly studied in the lab. To treat animal model organisms with fungi-derived molecules, extractions from different life cycle stages of fungi are performed. This includes mycelium (the vegetative stage), sporocarps (the reproductive stage), or spores (the end product of the reproductive stage), and each stage provides a variety of bioactive compounds. These bioactive compounds include glycoproteins, polysaccharides, triterpenoids, meroterpenoids, sesquiterpenoids, steroids, alkaloids, benzopyran derivatives, and benzoic acid derivatives, amongst others. In this work, we review evidence that fungal extracts from multiple species can have beneficial effects on the lifespan and healthspan of model organisms, such as *C. elegans* worms, *D. melanogaster* flies, and *M. musculus* mice. We cover extraction methods and lifespan effects of *Ganoderma lucidum* (i.e. Reishi), *Lentinula edodes* (i.e. Shiitake), the genus *Auricularia* (i.e. jelly ear mushrooms), the genera *Cordyceps* and *Ophiocordyceps* (e.g. the caterpillar fungi), *Hericium erinaceus* (i.e. Lion's mane), the mold genus *Monascus*, and also *Inonotus obliquus* (i.e. Chaga), *Grifola frondosa* (i.e. Maitake or hen-of-the-woods), the genus *Pleurotus* (e.g. oyster mushrooms), and *Agaricus subrufescens* (i.e. the almond mushroom). We propose that further research on fungal extracts will provide a greater picture of the role fungi-derived bioactive molecules can have in developing longevity and neuroprotective medicines for humans.

1. Methods

The following review is a narrative review (rather than systematic review) and includes studies that were qualitatively assessed for their relevance to the review topic. Briefly, studies were included in this review if they (i) described treatment of model organisms with a medicinal mushroom and (ii) looked at outcomes including lifespan, healthspan, or age-related changes. Studies that only focused on laboratory models of age-related diseases (e.g. cancer, diabetes, heart diseases, neurodegeneration) or oxidative stress, without addressing the natural aging process and measurement of lifespan change, were not necessarily included. Therefore, species of fungi that solely possessed evidence using non-natural aging models (e.g. oxidative stress cell culture models or D-galactose induced premature aging models) were not included. We note that these species may later have more solid literature evidence published in favor of their longevity effects. Medicinal mushrooms were defined as mushrooms that have been studied in a context to treat diseases, as outlined in Venturella et al. (2021), Kurtzman (2005), Wasser (2011). Databases and literature resources including PubMed, Google Scholar, and Mendeley were queried or searched using key words including "aging," "longevity," "lifespan," "healthspan," "antiaging", "fungi", "medicinal fungi", "mushroom", "medicinal mushroom", along with specific mushroom species names (from Venturella et al., 2021; Kurtzman, 2005; Wasser, 2011, or commonly accepted as medicinal), and combinations thereof. In certain cases, to be able to compare findings of different studies, we converted time-based values of lifespan

* Corresponding author at: Amsterdam UMC location University of Amsterdam, Laboratory Genetic Metabolic Diseases, Meibergdreef 9, Amsterdam, the Netherlands.
E-mail address: g.e.janssens@amsterdamumc.nl (G.E. Janssens).

https://doi.org/10.1016/j.arr.2023.101929
Received 29 December 2022; Received in revised form 5 April 2023; Accepted 6 April 2023
Available online 7 April 2023
1568-1637/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

change (e.g. "days") into percentages (comparing average treatment lifespan to average control lifespan). Standard deviations (SD) were not taken into consideration, since not all papers included SD values for the lifespan. Lifespan studies from the selected literature were done in model organisms including worms (Sutphin and Kaeberlein, 2009) flies (Linford et al., 2013) and mice (Flurkey et al., 2007). For detailed methods on how lifespan studies were conducted, the reader is direct towards the relevant reviews or each individual study's methods section Table 1.

2. Introduction

Aging is a complex process that happens in every living organism (Gems and Partridge, 2013). In humans it often involves the onset of the ineffective use of internal and external resources, producing intra- and extracellular damage and deregulation, resulting in the decrease or even loss of function of different tissues and organs (López-Otín et al., 2013). With this age-related loss of function comes age-related disease, and therefore, it is of great importance not only to fill in the gaps in our knowledge about aging, but also to continue to look for the molecules or combinations of compounds that can help us minimize detrimental symptoms people deal with when reaching old age. Studying the effects of compounds on lifespan and healthspan in model organisms allows us a window into such processes and interventions.

Both pharmaceuticals (i.e. medical compounds for treatment or prophylaxis of a disease) and nutraceuticals (i.e. a diet supplement that delivers a concentrated form of a biologically active component of food in a nonfood matrix to enhance health (Zeisel, 1999) represent potential candidates for health- and life-promoting molecules. To investigate a compound for such influence on the organism, molecular changes associated with longevity signaling pathways or mechanisms are often taken into consideration. This can include assessing the expression of stress resistance genes, where their upregulation is often associated with increased lifespan in model organisms (Lithgow and Walker, 2002). These can include genes such as the Forkhead Box O3 (FOXO3) transcription factor (DAF-16 in worms), the NFE2 Like BZIP Transcription Factor 2 (NRF2, known as SKN-1 in worms), the superoxide dismutase genes (known as the SODs across multiple species), sirtuins (including SIR-2.1 in worms) and the family of heat shock proteins (the HSPs, including HSP-16.2 in worms), as relevant for this review, amongst others.

Fungi were always given a great deal of attention and reverence in traditional medical practices, and now they have found their way into modern medicine, where mycelium, fruiting bodies, spores, and isolated fungi-derived molecules are being studied to clarify their place in medical practice. Extracts of these various fungal life stages (mycelium, primordium, fruiting bodies, spores), either by ethanol, hot water, or other polar and non-polar solvents, produce concentrated mixtures of primary and secondary metabolites from fungi, which often include glycoproteins, polysaccharides, triterpenoids, meroterpenoids, sesquiterpenoids, steroids, alkaloids, benzopyran derivatives, and benzoic acid derivatives. Indeed, fungi-derived polysaccharides have long been known to be of medicinal value (Shih et al., 2005; Lemieszek and Rzeski, 2012). Mycelium and fruiting bodies are usually studied for their polysaccharide content, meanwhile, spores contain many triterpenoids, that are considered to be responsible for the bioactivity of spore oil (Zhang et al., 2021). With abundant background knowledge on using sporocarps of fungi for promoting health in traditional medicine, and their emerging roles on longevity being scientifically explored in the lab, fungal extracts represent a rich source of bioactive molecules for the future development of medicines that promote healthspan and lifespan, termed 'longevity medicines' or 'geroprotectors'.

Multiple reviews exist highlighting the medicinal role of fungi as treatments for specific age-related diseases, including cancer (Moradali et al., 2007), neurodegeneration (Chaudhary et al., 2020), and metabolic disorders (Liang et al., 2013; Agrawal et al., 2022) along with others. For the role of fungi as treatments for these diseases the reader is referred to these comprehensive reviews, in addition to reviews on medicinal mushrooms in general (Wasser, 2011; Gründemann et al., 2020; Kurtzman, 2005). The current work will review the role of fungi as sources of longevity-promoting compounds. Fungi are included in this review if direct evidence exists that the lifespan of model organisms, including yeast, worms, flies, and mice, can be influenced by treatment with fungal extracts. Specifically, we cover fungi including *Ganoderma lucidum* (i.e. Reishi), *Lentinula edodes* (i.e. Shiitake), the genus *Auricularia* (i.e. jelly mushrooms), the genus *Cordyceps* (e.g. the caterpillar fungus), *Hericium erinaceus* (i.e. Lion's mane), the mold genus *Monascus*, *Inonotus obliquus* (i.e. chaga), *Grifola frondosa* (i.e. Maitake or hen-of-the-woods), genus *Pleurotus* (e.g. oyster mushrooms), and *Agaricus subrufescens* (i.e. the almond mushroom) (Fig. 1). These fungi cover a broad range of genera fulfilling diverse ecological roles, including lignicolous, saprobic, entomopathogenic, or molds, and demonstrate varying degrees of lifespan-prolonging effects (Fig. 1).

3. *Ganoderma lucidum*

In China, *Ganoderma lucidum* goes by the name of Lingzhi, in Japan by Reishi or Mannentake, and in Korea, this species is most often referred to as Youngzhi. *G. lucidum* could be considered to be one of the most well-studied and famous mushrooms for longevity, and has been nicknamed the so-called 'Mushroom of Immortality'. *G. lucidum* is a species of polypore mushrooms that can be found growing on live hardwood (primarily oak trees) as a parasite and on fallen hardwood as a saprobe. It has long been known to possess medicinal value and is still widely used in traditional Chinese, Japanese, and Korean medicine. Currently, *G. lucidum* is studied for a variety of activities: immunomodulatory, anti-tumor, hepatoprotective, antioxidant, anti-hyperglycemic, and antilipidemic (Wang et al., 2020). Health benefits of *G. lucidum* have been linked to high molecular weight polysaccharides that regulate healthy gut microbiota, and thus have an ameliorative effect on metabolic syndromes, producing anti-obesity, anti-diabetic and anti-inflammatory effects (Chang et al., 2015).

In one study (Cuong et al., 2019), a concentrated hot water extract from *G. lucidum* fruiting bodies (300 mg/mL) was used to perform a lifespan assay on *C. elegans*. Findings demonstrated that pretreatment with *G. lucidum* extract increased the worm's lifespan at the concentrations of 3.75 mg/mL and 7.5 mg/mL, while extract with a concentration of 15 mg/mL had the opposite effect. Compared with the control group, *G. lucidum* extract at a concentration of 7.5 mg/mL increased the median lifespan from 17.3 days to 23.5 days and maximum longevity from 34 days to 38 days. The optimum dose of 7.5 mg/mL *G. lucidum* extract was studied further and found to markedly reduce the body fat content and number of eggs laid, but significantly extended the laying period. In this study prolonged lifespan was observed in *daf-16*, *daf-2*, *eat-2* and *rsks-1* mutant worms, but not in *glp-1* mutants, which suggested that the effect of *G. lucidum* extract on longevity indeed depended on the gremlin signaling pathway.

Another study (Peng et al., 2021) focused on the hot water extract from the mycelium of *G. lucidum*. In this study the lifespan assay was performed using two concentrations of *G. lucidum* extract - 2 mg/plate and 20 mg/plate, several concentrations of *G. lucidum* subfractions in the range 0.1–2 mg/plate, and rapamycin to compare, dosed at 400 μM/plate. The results revealed that *G. lucidum* extract used at 2 mg/plate extended the median lifespan of *C. elegans* from 13.0 ± 2.8 to 18.5 ± 6.3 days compared to control water, representing a 45 % extension. In this case, the low *G. lucidum* concentration (2 mg/plate) was more effective than the high concentration (20 mg/plate), which provided a lifespan extension of 19 %, consistent with a hormetic dose response. Lifespan extension was not observed in *atg-18*, *daf-16,* and *sir-2.1* mutant worms, therefore the autophagy pathway was considered responsible for the life-prolonging effect. Along with this, *G. lucidum* also increased the expression of the LGG-1 protein, the ortholog of

Table 1
Summary of fungi and study results described in this review.

Latin	Common name	Extract or compound	Lifespan and/or healthspan effect	Tested organism	Pathway	Ref.
Ganoderma lucidum	Reishi, Lingzhi, Youngzhi, "Mushroom of Immortality"	Hot water extract from the fruiting body	Median lifespan extended by 35 % and maximum longevity extended by 11 %	*C. elegans*	Gremlin signalling pathway	Cuong et al. (2019)
		Hot water extract from mycelium	Median lifespan - 45 % extension; Maximum - 53 % extension	*C. elegans*	ATG-18, DAF-16, and SIR-2.1-dependent autophagy and stress resistance pathways	Peng et al. (2021)
		Ganodermasines A and B, extracted from spores	Extended lifespan	*S. cerevisiae*	UTH1	Weng et al. (2010)
		RF3 (Reishi polysaccharide fraction 3)	35 % lifespan extension RF3 alone RF3+ acetic acid - 74 % lifespan extension (the optimal concentration)	*C. elegans*	DAF-16 expression regulation via TIR-1 receptor and MAPK pathway	Chuang et al. (2009)
		GLSO (Ganoderma lucidum spore oil)	Mean lifespan - by 11.85 % in females and by 10.13 % in males Maximum lifespan - by 6.45 % in females and by 5.31 % in males	*D. melanogaster*	Antioxidant pathway through higher expression of SOD and CAT	Zhang et al. (2020)
		Combined water, absolute ethanol and petroleum ether extracts	Male: 80 mg/mL concentration increased average lifespan by 42.32 % and maximum lifespan by 4.49 % Female: 5 mg/mL concentration increased average lifespan by 27.4 % and maximum lifespan by 10.73 %	*D. melanogaster*	-	Huang et al. (2011)
		Triterpenoids, petroleum ether extracts from fruiting bodies	Regulation of sphingolipid metabolism, longer telomere length, and enhanced autophagy	Aged mice	-	Zeng et al. (2021)
Lentinula edodes	Shiitake, black forest mushroom, xianggu, golden oak mushroom	Combined water, absolute ethanol and petroleum ether extracts from fruiting body	Male: 5 mg/mL increased average lifespan by 40.53 % Female: 5 mg/mL increased average lifespan by 6.03 %	*D. melanogaster*	-	Huang et al. (2011)
		L2 - L. edodes-derived heteropolysaccharide	Decreased levels of following metabolites in urine: b-hydroxybutyrate, glutamate, glucose, choline, lactate Increased level of a-oxoglutarate in urine	Aged mice	-	Xiaofei et al. (2017)
		Crude polysaccharide extract from fruiting bodies	25 % mortality threshold in male flies: postponed by 14–20 % The lifespan of the first 10 % expiring male imagos: extended by 62–66 % The lifespan of the first 10 % expiring female imagos: shortened by 14–31 %	*D. melanogaster*	-	Matjuskova et al. (2014)
Auricularia auricula-judae	Jelly ear, Judas's ear, wood ear	AAPHs-F - acid hydrolysates of A. auricula-judae polysaccharides, NaOH extraction from fruiting bodies	Antioxidant (enhanced levels of SOD, CAT, GR, GSH, decreased levels of ROS and MDA) Maximum lifespan increased by 22.2 % at 0.5 mg/mL and by 11.1 % at 1 mg/mL concentrations of AAPHs-F	*C. elegans*	DAF-16, SKN-1, SIR, SOD-1 and SOD-2	Fang et al. (2019)
		Combined water, absolute ethanol and petroleum ether extracts from fruiting body	Male: 5 mg/mL increased average lifespan by 31.4 % and half death time by 33.7 % 20 mg/mL increased average lifespan by 28.9 % and half death time by 34.2 % Female: 20 mg/mL improved average lifespan by 16.9 %, half death time by 24.6 % and maximum lifespan by 10.5 %	*D. melanogaster*	-	Huang et al. (2011)

(*continued on next page*)

Table 1 (*continued*)

Latin	Common name	Extract or compound	Lifespan and/or healthspan effect	Tested organism	Pathway	Ref.
		AAPs (Auricularia auricula polysaccharides) obtained from NaOH extracts from fruiting bodies	Average lifespan: AAP1 - increased by 19.9 %, AAP3 - by 17.6 % Maximum lifespan: AAP1 - increased by 20 %, AAP2 and AAP3 - increased by 10 % Increased SOD and CAT activity	*C. elegans*	-	Xu et al. (2016)
Auricularia polytricha	Wood ear, cloud ear fungus	Ethanol, hexane and water extracts from fruiting bodies, but only ethanol one was tested for life-prolonging properties	Mean lifespan: increased by 4.3 % (20 μg/mL) and 16.2 % (40 μg/mL) Maximum lifespan: increased by 4 % (20 μg/mL) and 12 % (40 μg/mL) Increased pharyngeal pumping, neuroprotection from glutamate-induced neurotoxicity	*C. elegans*	Antioxidant (increased gene expression of SOD and GPx)	Sillapachaiyaporn et al. (2021)
*Ophiocordyceps sinensis (*form. *Cordyceps sinensis)*	Caterpillar fungus, "winter worm, summer grass"	CSOL (Cordyceps Sinensis Oral Liquid) – combined *C. sinensis* extract and mycelial fermentation broth	25 %, 32 % and 31 % lifespan extension at concentrations of 00.2, 00.6 and 0.20 mg/mL, respectively Inhibited accumulation of lipofuscin Increased activity of SOD1, CAT Mitigated effects H_2O_2 and paraquat-induced oxidative stress Mitigated effects of D-gal mimicked aging	*D. melanogaster*	-	Zou et al. (2015)
Cordyceps militaris	Scarlet Caterpillar club, Caterpillar fungus	CP2-c2-s2-polysaccharide, extracted from stroma	Increased lifespan by 16.6 % at the concentration of 150 μg/mL, and of 300 μg/mL - by ~ 10 %	*C. elegans*	-	Liu et al. (2016)
Grifola frondosa	Maitake, hen-of-the-woods, "dancing mushroom"	Hot water extract from the fruiting body	Median survival by 17.6 % higher after treatment with 20 μg/mL	*C. elegans*	SKN-1/NRF-2 and DAF-16/FOXO	Aranaz et al. (2021)
Hericium erinaceus	Lion's mane, Satyr's beard, Yambushitake, tooth fungus	Lyophilised erinacine A-enriched mycelium (5 mg/g)	***D. melanogaster*** Maximum lifespan increased by 14 %, 21.3 %, 17.6 % in male flies, by 16.7 %, 27.7 % and 31.9 % in female flies by 0.11, 0.35 and 1.05 mg/mL concentrations respectively Mean lifespan increased by 29.6 % in male flies and by 25.3 % in female. **Aged mice** Maximum lifespan increased by 23 % in male mice, by 14.3 % in female mice at the concentration of 431 mg/kg BW/day Mean lifespan increased by 20 % in both female and male mice at high-dose.	*D. melanogaster* and aged mice	-	Li et al. (2019)
		Ethanol extraction from dried primordium, ERGO-enriched mycelium	Locomotor frailty index decreased by 43.45 % Decreased immunoreactivity of COX2 and NOS2 Increased immunoreactivity of p35	Mice	-	Roda et al. (2022)
		Ethanol extraction from lyophilized mycelium and sporophore (fruiting body)	Decreased cognitive frailty index Increased PCNA positivity in cerebellar and hippocampal cells Increased DCX positivity in hippocampal cells	Frail mice (21.5 months of age)	PCNA/DCX	Ratto et al. (2019)
Inonotus obliquus	Chaga	Hot water extract from fruiting body	Increased mean lifespan by 22 %, 21 %, 21 % at concentrations of 50, 200,	*C. elegans*	-	Scerbak et al. (2016)

(*continued on next page*)

Table 1 (*continued*)

Latin	Common name	Extract or compound	Lifespan and/or healthspan effect	Tested organism	Pathway	Ref.
			800 μL/mL respectively Decreased number of abnormally, non-spontaneously moving and immobile nematodes Improved anterior touch response Altered mechanosensory neuron aging			
Monascus sp.	Red yeast	Monascin	Reduced amount of ROS Extended nematode's lifespan Delayed paralysis phenotype	*C. elegans*	Elevated gene expression of daf-16, sod-1, sod-2, sod-3, hsp-16.2 and prdx-2 DAF-16 is a key player	Shi et al. (2016)
		Monascin	Increased amount of TH+ cells, decreased amount of caspase-3 Inhibited expression of IL-1β, IL-6, TNF-α and phosphorylation of NF-kB Increased SOD and GPx activity, decreased levels of malondialdehyde	Rats	PPARy pathway	Lin et al. (2020)
*Agaricus subrufescens (*form. *A. blazei)*	Almond mushroom, almond portobello, mushroom of the sun, royal sun agaricus, jisongrong, himematsutake	Combined water, absolute ethanol and petroleum ether extracts from fruiting body	For the male flies, 5 mg/mL and 20 mg/mL exerted similar effect, extending average lifespan by 32 % and 30.5 % respectively, while 80 mg/mL decreased lifespan by 23 %. Male maximum lifespan was significantly higher only at 5 mg/mL, representing 9 % increase, and 80 mg/mL was again responsible for the decrease in maximum lifespan. Average female lifespan was not affected significantly by any of the three extract concentrations, but at 5 mg/mL maximum female lifespan was prolonged by 13.6 %, but at higher concentrations it dropped significantly.	*D. melanogaster*	-	Huang et al. (2011)
Pleurotus spp.	Oyster, abalone mushrooms or tree mushrooms	Dried fruiting bodies by lyophilization, and then ground to a fine powder	The lowest dose of the P. djamor ECS-0142 strain provided greater survival rates for the Mexican flies	*A.ludens*	-	Sánchez et al. (2015)
Not species-specific	Found across different mushroom species	L-ergothioneine	-	-	Regulation in cholinergic neurotransmission, tyrosine metabolism, FAO, and autophagy	Pan et al. (2022)

mammalian light chain 3 (LC3), a marker of autophagy. In conclusion, this study suggested that polysaccharides from *G. lucidum* could be considered prime candidates for CR mimetics since they are not digested by human digestive enzymes (they are poorly absorbed in the human digestive tract) and they may inhibit digestive enzymes and lipid absorption, but produce only minor side effects if any. These conclusions are in accordance with a recent review, suggesting a calorie-restriction effect and health-beneficial role of poly- and oligosaccharides in water extract of *G. lucidum* (Martel et al., 2021).

Not only fruiting bodies and mycelium of *G. lucidum* are valuable sources of potential anti-aging compounds, but its spores as well (Soccol et al., 2016; Guo et al., 2009). A study (Weng et al., 2010) investigated the anti-aging properties of two molecules extracted from *G. lucidum* spores - ganodermasines A and B. These compounds were extracted from spores using methanol extraction with consequent separation. The lifespan assay was performed on the K6001 yeast strain, and mutants of the UTH1 and SKN7 genes (genes ascribed to stress resistance and lifespan), all of which were grown on agar plates with different concentrations of the ganodermasines samples (1, 10, and 100 μM). The *lucidum*-derived ganodermasines extended yeast lifespan in a manner dependent on UTH1 and SKN7, suggesting activation of stress resistance pathways were responsible for the lifespan effects.

Another extract of *G. lucidum*, termed Reishi polysaccharide fraction 3 (RF3), was tested on *C. elegans*, as a separate fraction, along with acetic acid (AcOH) (Chuang et al., 2009). RF3 and acetic acid have different pathways when exerting life-prolonging effects, providing the ground for using these compounds separately and in combination. RF3 alone extended the nematode's lifespan by 35 %, and even though acetic acid had less of an impact on promoting lifespan, these substances together prolonged lifespan by 74 %. Not only the effect of this polysaccharide fraction was assayed, but also at different concentrations and whether storage time changed its properties. The most effective was a combination of AcOH 50 ppm and RF3 100 ppm, and this ratio was further tested after different storage times. It was observed that even after 2

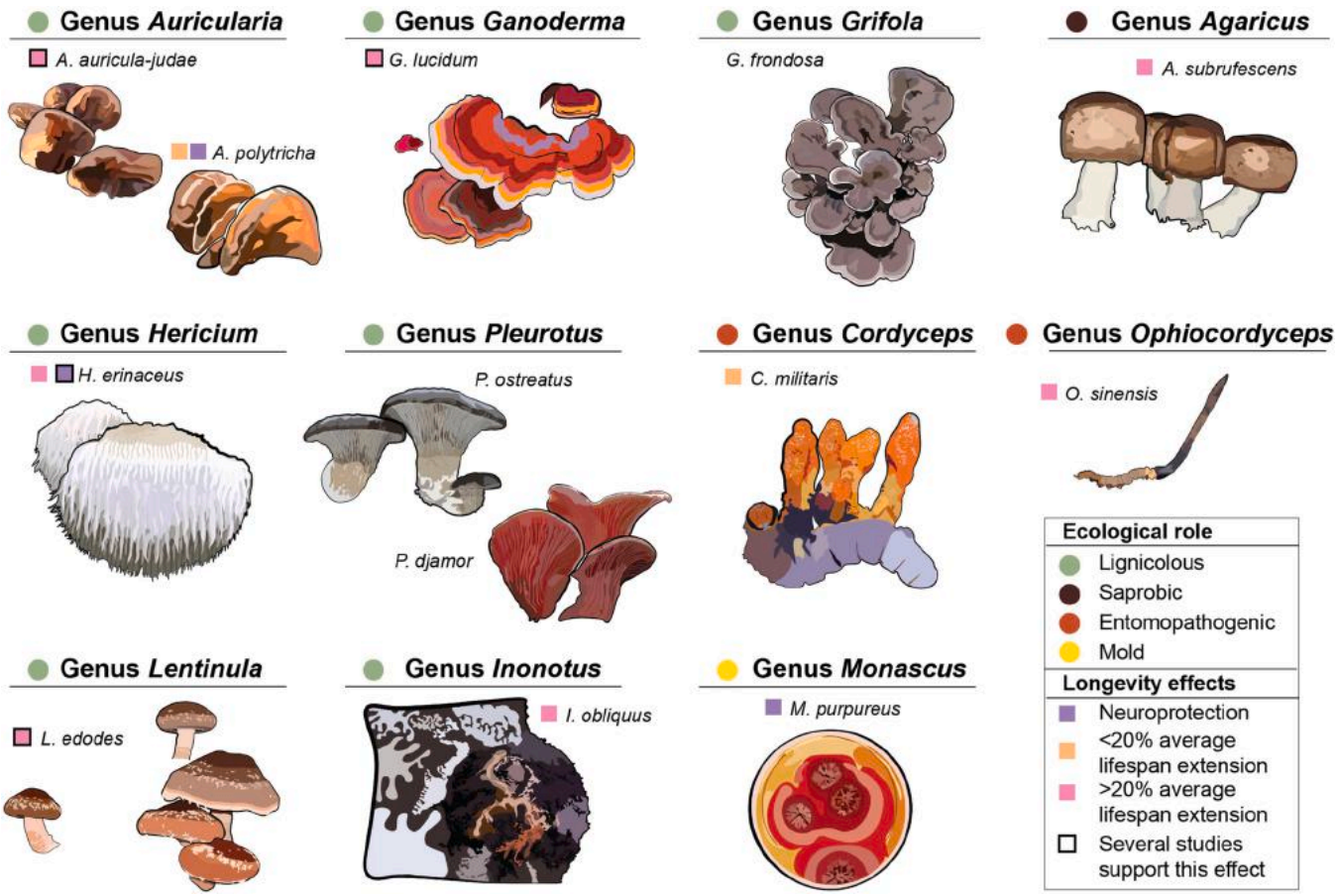

Fig. 1. Graphical representation of fungal species described in this review. Includes genus *Auricularia* (*A. auricula-judae* and *A. polytricha*), genus *Ganoderma* (*G. lucidum*), genus *Grifola* (*G. frondosa*), genus *Agaricus* (*A. subrufescens*), genus *Hericium* (*H. erinaceus*), genus *Pleurotus* (*P. ostreatus* and *P. djamor*), genus *Cordyceps* (*C. militaris*), genus *Ophiocordyceps* (*O. sinensis*), genus *Lentinula* (*L. edodes*), genus *Inonotus* (*I. obliquus*), and genus *Monascus* (*M. purpureus*). Ecological roles of fungal genera are denoted (lignicolous, saprobic, entomopathogenic, or mold), and longevity effects are summarized (> 20 % or < 20 % lifespan extension of a model organism) from the studies covered in the review.

weeks of storage, this combination was still active. The mechanism of this effect was studied extensively and it was determined that RF3 activates DAF-16 expression via the TIR-1 receptor and MAPK pathway, meanwhile acetic acid inhibits the trans-membrane receptor DAF-2 of the insulin/IGF-1 pathway to indirectly activate DAF-16 expression, indicating that DAF-16 is a central gene involved in this life-prolonging effect.

Another study (Zeng et al., 2021) focused on determining whether triterpenoids from *G. lucidum* (TGL) can delay or mitigate age-associated changes in various organs, specifically in the brains of mice. To assess *G. lucidum*-induced changes with regard to mitigating physiological decline in organ functions that comes with age, long-term administration of triterpenoids via dietary supplementation was carried out. The study demonstrated that after the administration of TGL, the telomere length in hepatocytes was found to be longer, the brown adipose tissue levels around the scapula were significantly improved, the level of senescence marker β-galactosidase in kidney tissue was reduced and ferroptosis, which leads to accumulation of iron in toxic levels, was inhibited. In addition to these effects, it was demonstrated that TGL-treated groups had fewer apoptotic cells in the brain tissue, the telomere length in brain tissue samples was longer in the female TGL-treated group, and the expression of phosphorylated-mTOR and LC3A/B were upregulated. Furthermore, sphingolipid metabolism and arachidonic acid metabolism were the most influenced pathways in the TGL-treated groups. Together this suggests triterpenoids from *G. lucidum* may delay (brain) aging by mitigating telomere shortening and activating autophagy possibly through sphingolipid and fatty acid metabolism.

From *G. lucidum* spores one can obtain not only ganodermasines A and B, as mentioned above (Weng et al., 2010) but also oil with a high content of triglycerides. In the study carried out by Zhang et al. (2021) *G. lucidum* spore oil (GLSO) was chosen to test its potential life-prolonging activity on fruit flies (*D. melanogaster*). The assay was performed with GLSO of different concentrations (low dose – 0.3125 mg/mL, medium dose – 0.625 mg/mL, and high dose – 1.25 mg/mL) and it was carried out under normal conditions and under the conditions of oxidative stress caused by hydrogen peroxide. The highest dosage of GLSO (1.25 mg/mL) increased lifespan of fruit flies significantly, mean lifespan by 11.9 % in females and by 10.1 % in males, and the maximum lifespan was increased by 6.5 % in females and by 5.3 % in males, relative to the control group. In order to investigate the molecular mechanisms underlying this effect, the influence of GLSO on the expression of Cu, Zn-SOD, Mn-SOD and CAT mRNA was evaluated. The mRNA expression of flies treated with 1.25 mg/mL GLSO was significantly increased in comparison to the control group: Cu, Zn-SOD by 68 %, Mn-SOD by 50 %, CAT by 83 % in females; Cu, Zn-SOD by 64 %, Mn-SOD by 66 %, CAT by 79 % in males.

In another study on Reishi (Huang et al., 2011) extraction was performed using *G. lucidum* fruiting bodies, successively soaking them in water, absolute ethanol, and petroleum ether. For the lifespan-assay, a mixture of the 3 aforementioned extracts was used. To perform *D. melanogaster* survival assays, basic media and media with 3 different concentrations of the extract (5, 20, and 80 mg/mL) were prepared. Grouping results by the sex of the fruit flies, average lifespan extension

for male flies was registered at the concentrations of 20 and 80 mg/mL of media, but maximum lifespan increased only in the group that was grown in media with 80 mg/mL of the extract. This concentration, 80 mg/mL, increased the average lifespan by 42.3 % and maximum lifespan by 4.5 %, implying that 80 mg/mL of *G. lucidum* extract was the optimal concentration in terms of extending the lifetime of male flies. As for the female flies, extracts at 80 and 5 mg/mL significantly increased the average lifespan, the latter also increased the maximum lifespan by 10.7 %.

4. *Lentinula edodes*

Lentinula edodes, also known as shiitake, is both an edible macromycete of high nutritional value and a medicinal mushroom. It is a xylotrophic white-rot-forming species that can be found in situ growing on oak, poplar, sugar maple, and birch trees, and for mass production, it is most often cultivated on wood chips and sawdust. Medicinal components of *L. edodes* include polysaccharides (e.g. Lentinan, L-II, heterogalactan, Lew, Lea), terpenoids, sterols, lipids and other bioactive molecules, such as eritadenine (Xu et al., 2014; Finimundy et al., 2014). Extracts from fruiting bodies and mycelium have shown an immunomodulatory effect, and anti-tumor, hepatoprotective and antioxidant activities, which are attributed mostly to its polysaccharide content (Chen et al., 2020; Sasidharan et al., 2010; Chen, Ju, et al., 2012). This is also one of the few mushrooms with reports on anti-caries activity, its extracts inhibited the adherence capability of *Streptococcus* species, one of the caries-inducing genera, suppressed the formation of biofilm and enhanced its disruption (Ponnusamy et al., 2022).

In the same article mentioned above for *G. lucidum* testing fungal extracts for effects on lifespan in flies (Huang et al., 2011), *L. edodes* was also assessed. The results revealed that *L. edodes* extract in concentrations of 5 mg/mL and 20 mg/mL exhibited similar life-prolonging effects. More specifically 5 mg/mL resulted in an average lifespan lengthening by 40.53 % for male and 6.03 % for female flies, meanwhile the maximum lifespan was not affected (or decreased insignificantly). However, as the extract concentration increased, the maximum lifespan of the fruit flies decreased and the average lifespan weakened, indicating that the lower dose of bioactive extract from *L. edodes* is the most effective in extending the lifespan of *D. melanogaster*.

Just as with *G. lucidum*, polysaccharides from *L. edodes* are considered to be responsible for the medicinal properties of this mushroom. One study (Xiaofei et al., 2017) investigated the influence of *Lentinula edodes*-derived heteropolysaccharide L2 on aged mice, using the 1H NMR-based metabolomics approach. In this study, six metabolites were identified as potential biomarkers in urine for L2 treatment: beta-hydroxybutyrate, glutamate, glucose, choline, lactate, the levels of which were decreased in L2-treated aged mice and alpha-oxoglutarate (a-OG), concentration of which was elevated after L2-treatment. Decreased level of glucose may indicate increased glucose utilization in the organism via glycolysis, the decreased level of beta-hydroxybutyrate in urine suggests a reduction of energy production from lipid oxidation, and the changed concentration of choline in urine indicates that L2 might play a role in central nervous system function, since it is critical for brain function and cognitive performance, being a precursor for acetylcholine. The decrease in lactate in urine, the end-product of glycolysis, suggests the decrease of anaerobic glycolysis in cells. And, lastly, increased levels of alpha-oxoglutarate may point to increased ATP generation in cells. The results of this study revealed that *L. edodes*-derived polysaccharides (L2) might play a role in mitigating the aging processes partially through the regulation of energy metabolism in aged mice.

Another study on polysaccharides from *L. edodes* described extraction using hot water, consequent ethanol precipitation, and drying for testing in *D. melanogaster* flies (Matjuskova et al., 2014). Before usage in lifespan assays, hot water extract (HWE) of polysaccharides was dissolved in distilled water to a final concentration of 20 mg/mL. Flies were fed with supplementation of shiitake HWE in different concentrations of polysaccharide fraction (PF) – 0.003 %, 0.007 %, 0.015 %, 0.03 %, 0.06 %, and 0.1 %. In general, it was found that males tended to have lifespan benefits from HWE, and of two doses used to assess activity levels with aging, a measure of healthspan, benefits occurred more frequently with the lower dose tested (0.015 %). Contrasting this, females fed HWE had in general decreased lifespan and healthspan with HWE extract. With the feeding and weight of flies unaltered with HWE treatment, it was hypothesized that the lifespan changes observed were not due to a calorie restriction effect.

5. Genus *Auricularia*

Auricularia is a genus of edible macrofungi, often called jelly mushrooms, that can usually be found growing on trees and shrubs. For a long time, species of this genus have been used in the culinary field, but during the last decades several species, including *A. auriculus-judae* and *A. polytricha*, have drawn the attention of scientists as sources of medicinal compounds. Just as with many fungi, the main bioactive molecules within this genus are polysaccharides (Glucan I, Glucan II, APPS-1), melanin, polyphenolic compounds, exo-biopolymer, and proteins. These compounds can have various effects including anti-viral (Nguyen, Chen, et al., 2012), immunomodulatory (Nguyen, Wang, et al., 2012), cytokine (IL-6, IL-10, TNF- α)-modulatory (Zhang et al., 2018), antithrombotic (Yoon et al., 2003) and antioxidant (Xu et al., 2016). Considering recent findings on prevalent hyperlipidemia in elderly patients (Rosada et al., 2020), anti-hyperlipidemic agents are gaining the spotlight in aging research. Extract and polysaccharides from *Auricularia species.* could also fill this role, exerting anti-hyperlipidemic effects, and reducing the level of total cholesterol and low-density lipoproteins (Zhang et al., 2020; Zhao et al., 2015).

5.1. *Auricularia auricula-judae*

Auricularia auricula-judae, more commonly known as jelly ear, is a brown, gelatinous, and ear-like shaped mushroom that grows on wood. In the same article described above (Huang et al., 2011) concerning *G. lucidum* and *L. edodes*, *A. auricula-judae* was also assessed for lifespan-prolonging activity in flies. At the concentrations of 5 mg/mL and 20 mg/mL, the average lifespan of the male flies was extended by 31.4 % and 28.9 %. Maximum lifespan was either influenced slightly or even decreased, which suggests that concentrations between 5 and 20 mg/mL may be the most suitable to prolong the lifespan of male *D. melanogaster*. As for the female flies, 20 mg/mL was the optimal concentration for prolonging the average lifespan and maximum lifespan, which were increased by 16.9 % and 10.5 % respectively. Concentrations of 5 and 80 mg/mL had also life-prolonging effects; 80 mg/mL was the second-best in increasing average lifespan, while 5 mg/mL was the second-most optimal concentration for maximum lifespan improvement.

Another study (Fang et al., 2019) performed in *C. elegans* worms, focused on researching life-prolonging and antioxidant effects of the acid hydrolysates of the polysaccharide fraction obtained from sodium hydroxide (NaOH) extract from *A. auricula-judae* fruiting bodies (termed APPHs-F). This extract was assayed for antioxidant activity in vivo, which included registering a change in lifespan, gene expression, and cell protection from apoptosis. At the concentration of 0,4 mg/mL the activity of anti-oxidative enzymes, such as SOD, CAT, and glutathione reductase (GR) was enhanced by 70.6 %, 73.5 %, and an astounding 258.7 %, respectively. In addition to that, GSH level increased by 110.2 %, while levels of MDA and ROS decreased by 46.2 % and 31.9 % respectively. The results of the lifespan assay demonstrated, that even though the mean lifespan was either not affected or even decreased, the maximum lifespan was increased by 22.2 % at APPHs-F concentration of 0.5 mg/mL and by 11.1 % at the concentration of 1 mg/mL. Several concentrations of APPHs-F (0.05, 0.1, 0.2, 0.4 mg/mL) were used to

check if it possessed any influence on the expression of longevity and stress-related genes (DAF-16, SOD-1, SIR-2.1, SKN1, and SOD2). The results showed, that concentration of 0.1 mg/mL upregulated the expression of all of these genes, which suggests that the antioxidant and life-prolonging effect of this polysaccharide fraction may be regulated through them. Lastly, all of the concentrations mentioned above had protective effects on cells against apoptosis.

Different kinds of polysaccharides from *A. auricularia-judae* have also been studied for antioxidant and life-prolonging properties. In another study (Xu et al., 2016), five polysaccharides (APP1-APP5) were obtained from five *A. auricularia-judae* varieties through an alkaline extraction process (NaOH) and tested in vitro and in vivo on *C. elegans*. In vitro assays demonstrated that all AAPs exhibited certain antioxidant activity, dose-dependent free radicals, and superoxide anion-scavenging activities. In vivo, these extracts exhibited life-prolonging effects on nematodes, i.e. treatment with AAP1 increased average worm lifespan by 19.9 %, and with AAP3 by 17.6 %. AAP2, AAP4, and AAP5 improved average lifespan only slightly. As for the maximum lifespan, AAP1 improved it by 20 % and AAP3, same as AAP2, increased it by 10 %, and AAP4 and AAP5 did not exert any influence. Among all 5 polysaccharides, AAP1 exhibited the highest protective property from paraquat and H_2O_2-induced oxidative stress and was able to increase the activities of SOD and CAT.

5.2. *Auricularia polytricha*

As for the other *Auricularia* species in this review, one study (Sillapachaiyaporn et al., 2021) focused on studying antioxidant, neuroprotective, and life-promoting properties of three different crude extracts from *A. polytricha* in the model organism *C. elegans*. Extracts were prepared from fruiting bodies, macerating them in hexane, ethanol, and water, and obtaining APH (hexane), APE (ethanol), and APW (water) extracts, respectively. Antioxidant and glutamate-neurotoxicity assays have shown, that APE was the most potent extract, asserting a neuroprotective effect in a dose-dependent manner and upregulating the expression of genes that encode antioxidant enzymes (SOD and GPx). Therefore, the same ethanol extract was chosen to test its ability to prolong the lifespan and healthspan of *C. elegans*. The results revealed, that at the concentration of 20 μg/mL the mean lifespan was prolonged by 4.3 % and the maximum lifespan was increased by 4 %, meanwhile, at the concentration of 40 μg/mL there was an increase of 16.2 % and 12 % correspondingly. Pharyngeal pumping rates and spawning were also considered, when studying the effect of APE, and it was demonstrated that APE, indeed, increased pharyngeal pumping rates, improving nematode's healthspan and that it did not affect *C. elegans* spawning.

6. *Cordyceps militaris*

Cordyceps is one of the most studied genera in the *Ascomycota* phylum, and its species fulfill their ecological role as entomopathogens (parasites of insects and other arthropods). Before 2007 Cordyceps was considered the most diverse genus within *Clavicipitaceae* family, but molecular phylogenetic analysis allowed researchers to divide species, previously belonging to Cordyceps genus, into four genera: *Ophiocordyceps*, *Cordyceps*, *Elaphocordyceps* and *Metacordyceps* (Sung et al., 2007). Because *Cordyceps* species have evolved to invade arthropods' organisms, evading the immune system and harmonizing with their life cycle to survive and propagate, it has led to the production of unique bioactive secondary metabolites, such as cordycepin and cordycepic acid. However, studies have attributed the medicinal properties of *Cordyceps* species not only to secondary metabolites, but also primary ones, such as polysaccharides, polyunsaturated fatty acids and amino acids.

C. militaris is one of the most medicinally outstanding species within the *Cordyceps* genus, which was not reclassified into genus after molecular phylogenetic analysis. It was and is still valued as a folk medicine in Asian countries and considered a cheaper substitute for *O. sinensis*. Cordycepin was isolated from this species, as well as other bioactive molecules – ergosterol, mannitol, and several polysaccharides. In addition to the biological activities shared within the genera Cordyceps and Ophiocordyceps (antioxidant, antitumor, hepatoprotective, immunomodulatory), *C. militaris* has been shown to possess anti-allergic, anti-HCV, HIV1 protease inhibiting, and anti-obesity activities (Niwa et al., 2013; Wu et al., 2021).

In the article (Liu et al., 2016) a specific polysaccharide from *C. militaris* was studied. Crude polysaccharides (CP) were extracted from *C. militaris* through sequential hot water extraction, ethanol precipitation, and zinc sulphate deproteination. To obtain a specific polysaccharide, CP2-c2-s2, the CP was fractionated using membrane separation and purified via chromatography. To analyze the medicinal properties of CP2-c2-s2, lymphocyte proliferation and lifespan assays were performed on *C. elegans*. At concentrations from 12.5 μg/mL to 50 mg/mL CP2-c2-s2 remarkably enhanced the proliferative activity of T and B lymphocytes, at the concentration of 50 μg/mL it had a better immunostimulatory effect than lentinan (previously described polysaccharide from *L. edodes*) of the same concentration. It is worth noting that, although having immunomodulatory activity at previously mentioned concentrations, T and B lymphocyte proliferation decreased as the concentrations increased from 100 μg/mL to 250 μg/mL. Administering CP2-c2-s2 at the concentration of 150 μg/mL increased lifespan by 16.6 %, CP2-c2-s2 and lentinan at the concentration of 300 μg/mL – by ~ 10 % and 10.5 % respectively. To investigate the anti-aging effect of CP2-c2-s2 further, changes in body movement, pharyngeal pumping, and intestinal lipofuscinosis were assessed. The results were as follows: CP2-c2-s2 significantly slowed down the decrease in motility, at the concentration of 150 μg/mL, CP2-c2-s2 delayed the decrease in pharyngeal pumping rates and lipofuscin formation to some extent. Fecundity was not affected by CP2-c2-s2.

7. *Ophiocordyceps sinensis*

O. sinensis, also known as the caterpillar fungus, is the most studied fungi within the newly formed genus *Ophiocordyceps*, species of which were previously classified within the *Cordyceps* genus. Its chemical constituents include a wide variety of bioactive polysaccharides, cordycepic acid, cyclic peptides, and sterols, although cordycepin was found in this species only in trace amounts, unlike in *C. militaris* (Yuan et al., 2022). Nonetheless, both in vivo and in vitro studies have confirmed its anti-tumor, antioxidant, anti-inflammatory, immunomodulatory, anti-diabetic, anti-atherosclerotic, hepatoprotective, and anti-fatigue biological activities (Nagata et al., 2006; Wang et al., 2012; Zhang et al., 2005).

One study investigated the effect of *O. sinensis* oral liquid (CSOL), prepared from *O. sinensis* extract, that was combined with *O. sinensis* mycelial broth on *D. melanogaster* lifespan with regards to oxidative stress relief (Zou et al., 2015). The lifespan assay was performed under normal conditions, conditions of H_2O_2 and paraquat-induced oxidative stress, and a D-galactose (D-gal) induced model of aging. CSOL was able to significantly increase the lifespan of the male fruit flies under normal conditions, resulting in 25 %, 32 %, and 31 % lifespan extension at concentrations of 0.02, 0.06, and 0.20 mg/mL, respectively. Along with that, CSOL also inhibited the accumulation of lipofuscin and increased the activity of SOD1 and CAT, although it did not affect SOD2 activity and the transcription of the respective genes was not increased. Under H_2O_2-induced stress conditions, CSOL was able to improve lifespan in a dose-dependent manner, ranging from 3 % to 16 %. In regards to the paraquat-induced oxidative stress, the most effective concentration of CSOL was 0.06 mg/mL, which resulted in a 24 % lifespan extension. With D-gal mimicked aging, CSOL was able to partially restore the lifespan and activity of SOD1, and CAT enzymes, while inhibiting lipofuscin accumulation.

8. *Hericium erinaceus*

H. erinaceus, also known as Lion's mane or Yambushitake, is one of the most well-known and commercialized species of medicinal fungi. It is a white-rot-causing saprobic species, but since it often can be found not only growing on dead trees but also on dead parts of living trees, it may sometimes be misidentified as parasitic (Boddy et al., 2011). Lion's mane is one of the few species of fungi that are being studied in current human clinical trials for its neuroprotective and cognitive improvement effects (Saitsu et al., 2019; Mori et al., 2009).

Erinacine A is a diterpene isolated from *Hericium erinaceus* widely studied for its neuroprotective and nerve-growth-stimulating properties (Bailly and Gao, 2020; Shimbo et al., 2005). One study (Li et al., 2019) focused on finding the influence of Erinacine A-enriched mycelia of *H. erinaceus* on the longevity of *D. melanogaster* and aged mice (*Mus musculus*). Erinacine A concentration in mycelia was quantified to be 5 mg/g. In a *D. melanogaster* survival assay, mycelium was added to the media in 3 concentrations: 0.11, 0.35, and 1.05 mg/mL, which were chosen on the grounds of the previous study with a risk of minimal toxicity (Li et al., 2014). The maximum lifespan of the male flies was prolonged by 14 %, 21.3 %, and 17.6 % (corresponding concentrations were 0.11, 0.35, and 1.05 mg/mL), meanwhile, the maximum lifespan of the female flies was extended in a dose-dependent manner - concentrations of 0.11, 0.35 and 1.05 mg/mL were responsible for 16.7 %, 27.7 % and 31.9 % lifespan increase correspondingly. As for the mean lifespan, for male flies it increased along with the Erinacine A concentration, reaching 36.7 days at a maximum concentration (1.05 mg/mL), which corresponds to a 29.6 % extension. The maximum prolonging effect on the mean lifespan of the female flies – 25.3 % - was exerted by 0.35 mg/mL of Erinacine A. In survival assays of the Senescence Accelerated Mouse-Prone 8 (SAMP8), a naturally occurring mouse line of accelerated aging, 3 concentrations of Erinacine A-enriched mycelium (108, 215, 431 mg/kg BW/day) were administered to mice by oral gavage. In comparison with the control, the maximum lifespan of the male mice was increased by 3 months (23 %) by the high-dose Erinacine A, while the maximum female lifespan went from 14 to 16 months at the same dose, indicating a 14.3 % increase in lifespan. The mean lifespan increased from 10 to 12 months, indicating a 20 % increase for both male and female mice. Although mechanisms underlying such life-prolonging influence of Erinacine A-enriched mycelium are not fully understood yet, this study also analyzed the change in oxidative stress parameters. Dose-dependent decrease of thiobarbituric acid reactive substances (TBARs; formed as a byproduct of lipid peroxidation) in liver tissue, along with the increased activity of antioxidant enzymes (SOD, CAT, GPx) could be partially responsible for the life-extending effect of *H. erinaceus*.

Another compound that was studied for life-promoting properties is L-ergothioneine (EGT), a thiol derivative of histidine. It is known to be synthesized mainly by mycobacteria, cyanobacteria, and non-yeast fungi (Cheah and Halliwell, 2012). A high content of EGT was found in either mycelium or the fruiting body of *C. militaris*, *C. cicadae*, *H. erinaceus*, *G. frondosa*, *A. bisporus*, though the most EGT-rich sources were found to be the fruiting bodies of *Pleurotus* species (Chen, Ho, et al., 2012). One study (Roda et al., 2022) investigated if EGT derived from primordium of *H. erinaceus* could improve locomotor performance in mice during aging and if there was any change in oxidative stress pathway markers (COX2, NOS2). Locomotor performance was evaluated using 4 locomotor parameters: resting time, total distance, maximum, and mean speed. Eight months of supplementation with *H. erinaceus* primordium extract, from 15 to 23 months of age, significantly decreased locomotor frailty index (43.45 % decrease), mainly attributed to changes in resting time and total distance.

A study evaluating the role of EGT in longevity used the isolated molecule in different concentrations conducted on two wild-type laboratory strains of *D. melanogaster* (Pan et al., 2022). For the *Canton-S* strain of flies, the female lifespan was extended the most, by 15 % at 100 μL, and for males by 12 % at 1000 μL. For *yw* strain of flies, the largest lifespan prolongation effect for female flies was 100 μL, resulting in a 16 % lifespan extension, with the second best being 1000 μL, which led to an 11 % lifespan extension. As for the male flies of the same strain, at the concentrations of 10 μL and 100 μL the lifespan was prolonged by 6 % and 7 % accordingly. Going into this effect deeper, these results were not observed in flies treated with antibiotics, which means that the lifespan extension effect is expressed in a gut-bacteria-dependent manner. Additionally, unlike many compounds and extracts that prolong the lifespan in *D. melanogaster*, EGT had no negative impact on fecundity. The physiological effect of L-ergothioneine included improving the climbing ability of aging flies (days 30 and 50), EGT was also able to help female flies maintain the same bodyweight and reduce TAG levels, which may point to a regulatory effect of fat storage in aged flies. The upregulation of the choline transporter and increased levels of acetylcholinesterase in the brains of flies supported the idea of EGT-evoked neuroprotective effects.

Metabolites—including Herinacine A, Hericenone C and Hericenone D—were obtained from the ethanolic extract of *H. erinaceus* (He1), and tested for effects of memory improvement and induction of neurogenesis in frail mice (Ratto et al., 2019). Several frailty indexes (FIs), Locomotor, Cognitive, and both Locomotor and Cognitive (LAC), were introduced to evaluate the functional decline that accompanies physiological aging. Supplementation with He1 from 21.5 months to 23.5 months of age decreased Cognitive FI from 1.71 to 0.72, but no influence on Locomotor FI was observed. To investigate this effect further, immunocytochemical studies were conducted on cerebellar and hippocampal cells, and proliferating cell nuclear antigen (PCNA) and doublecortin (DCX) were used as specific markers for active proliferation and neurogenesis, respectively. Quantitative analysis showed that PCNA labeling frequency was higher in both hippocampus and cerebellum (from 10.80 % to 22.89 % and from 8.19 % to 25.60 % respectively). DCX labeling was significantly higher in the hippocampus after He1 treatment, and a trend for an increase of DCX positivity in the cerebellum was observed, but it did not reach the statistically significant threshold. Both of these markers can point to neurogenesis, induced by oral supplementation with *H. erinaceus* extract, more specifically PCNA positivity is connected with increased DNA repair and duplication, while DCX positivity is linked with the presence of newborn neurons.

9. Genus *Monascus*

Monascin is a yellow pigment produced by species of the *Monascus* genus. The most well-known species of this genus are *M. purpureus*, *M. ruber* and *M. pilosus*, which have widespread usage in red-yeast (also known as *Monascus)* fermentation of foods, mainly of rice and dioscorea. This compound has been used as medicine in China for ages and recent studies have confirmed its hypolipidemic, high-density cholesterol-raising, anti-inflammation, neuroprotective and antioxidant properties (Shi and Chen, 2020; Lee et al., 2010; Lai et al., 2021). One study investigated monascin as an antioxidant agent, its ability to reduce amyloid-β toxicity that accompanies Alzheimer's disease, and its effects on the lifespan of *C. elegans*. It was demonstrated, that at a concentration of 1 μM, it was able to reduce the amount of ROS, extend worm lifespan, delay paralysis phenotype in worm models that accumulate amyloid-β and even reduce amyloid-β levels in nematodes that have already accumulated it (Shi et al., 2016). To elucidate the mechanism underlying this effect, real-time PCR analysis and RNAi assays were performed. It was shown that this effect was accompanied by elevated gene expression of *daf-16, sod-1, sod-2, sod-3, hsp-16.2,* and *prdx-2* and that DAF-16 is a key player in this neuroprotective effect against amyloid-β plaques formed during Alzheimer's disease progression.

10. *Inonotus obliquus*

Inonotus obliquus, also known as Chaga, is a parasitic fungi species

that often can be found growing on birch trees. It is also a known traditional medicine agent, that was used to treat gastrointestinal diseases (Szychowski et al., 2021). Studies have demonstrated that it possesses anti-inflammatory, immunomodulatory, antioxidant, antiviral, anti-cancer, and even hypoglycemic effects (Sim et al., 2016; Chen et al., 2015; Liu et al., 2018). One study investigated *I. obliquus*, an Alaskan species used in botanical treatments, in terms of extending lifespan, healthspan, and altering mechanosensory neuron aging in *C. elegans* (Scerbak et al., 2016). Crude extract of *I. obliquus* was prepared using hot water, to imitate the tea-like infusion that is often consumed. Several concentrations, 50, 200, and 800 μL/mL of extract prolonged mean lifespan by 22 %, 21 %, and 21 % accordingly. It is worth noting that this effect was not observed when *C. elegans*'s diet consisted of UV-killed bacteria, suggesting fungi extracts might affect *C. elegans* through changes in *E. coli* metabolism. Several measures of healthspan were taken to elucidate life-prolonging effects and whether it contributes to the mitigation of the frail state that comes with aging. Treatments with Chaga extract helped nematodes maintain healthy mobility well into adulthood. On top of that, the highest concentration of *I. obliquus* extract (800 μL/mL) also decreased the incidence of immobile nematodes. Lowest and medium doses of Chaga extract also improved anterior touch response, in comparison to controls, although no difference in posterior touch response was present. ROS levels were significantly decreased in the *I. obliquus* nematode group, although another marker of oxidative stress – HSP-16.2 (Heat Shock Protein-16.2) was not affected.

11. *Grifola frondosa*

Grifola frondosa, or Maitake or hen-of-the-woods, is as much of a culinary mushroom as it is medicinal. It provides a wide array of compounds, from β-glucans and heteroglucans to proteins and glycoproteins, and both of the groups are being assayed for anti-tumor and immunomodulatory activities (Fang et al., 2012; Seo et al., 2019; Yuan et al., 2019). In one study (Aranaz et al., 2021) *G. frondosa* fruiting bodies were used for hot water extraction. This extract was studied using *C. elegans* for antioxidant, genotoxic, lipid-homeostatic, and life-promoting properties. To do in vivo assays, the genotoxicity of *G. frondosa* extract (GE) was evaluated, and it was shown to be non-toxic. The results of the in vivo assays demonstrated that GE significantly reduced lipid accumulation in worms, without affecting worm development, and further investigation of the mechanism responsible for this has led researchers to believe that DAF-16-FOXO signaling pathway was underlying this effect. Treatment with GE also induced a significant reduction in ROS in *C. elegans* and it was established that this activity was mediated by the skn-1/Nrf-2 transcription factors. Even though lifespan was not affected by GE at the concentration of 10 μg/mL, 20 μg/mL proved to increase median survival by 17,6 % and induced a significant reduction of age-related pigment lipofuscin.

12. Genus *Pleurotus*

Pleurotus is considered to be mostly a genus of edible mushrooms of high nutritional and culinary value, comprising the well-known 'oyster mushrooms', but it is also used in bioremediation, degradation of xenobiotics, enzyme production, bioconversion of lignocellulosic wastes, recycling agricultural residues, and, most interestingly, it is being uncovered now as a mushroom of medicinal value (Singh et al., 2012). *P. ostreatus* (pearl oyster), *P. djamor* (pink oyster), *P. pulmonarius* (lung oyster) and *P. eryngii* (king oyster) are the most popular species within this genus that are being studied in medical fields for anti-tumor, antioxidant and antiatherosclerotic activities (Krakowska et al., 2020; Maity et al., 2021; Singh et al., 2012). In one study (Sánchez et al., 2015) the influence of two *Pleurotus* species, *P. djamor* and *P. ostreatus* on longevity of the fly species *A. ludens* was not as unequivocal. After evaluation of antioxidant capacity, two fungal strains (ECS-0142, ECS-1123) of two species (*P. djamor* and *P. ostreatus,* respectively) in three concentrations (20 %, 5 %, 1 %) were chosen to evaluate the influence on Mexican fruit fly longevity, where *P. djamor* ECS-0142 is a strain with high antioxidant capacity, *P. ostreatus* ECS-1123 showed low antioxidant capacity. The highest concentration of *P. ostreatus* ECS-1123 impacted the lifespan of the flies negatively, resulting in the lowest life expectancy (decrease by 23.3 % for females, by 23.5 % for males) and higher mortality rates, while the lowest concentration of *P. djamor* ECS-0142 was the only treatment that resulted in significantly greater survival for both sexes. Decreased lifespan expectancy could potentially be explained by high protein content, which is known to adversely affect longevity.

13. *Agaricus subrufescens*

Agaricus subrufescens, also known as almond mushroom, mushroom of the sun, God's mushroom, or mushroom of life, is a saprobic edible species, that can be found growing on litter or rich soil, and is a famous cultivated species for culinary purposes. Not only is it an edible species of high culinary value, but it also possesses anti-tumor and antioxidant activities (Soares et al., 2009; Takaku et al., 2001). The fourth species of the study, described in detail above, that investigated the life-prolonging effect of four edible fungi species (Huang et al., 2011) was *A. subrufescens.* For the male flies, 5 mg/mL and 20 mg/mL exerted a similar effect, extending average lifespan by 32 % and 30.5 % respectively, while 80 mg/mL decreased lifespan by 23 %. Male maximum lifespan was significantly higher only at 5 mg/mL, representing a 9 % increase, and 80 mg/mL was again responsible for the decrease in maximum lifespan. The average female lifespan was not affected significantly by any of the three extract concentrations, but at 5 mg/mL maximum female lifespan was prolonged by 13.6 %, though at higher concentrations it dropped significantly.

14. Discussion

In this review, we found evidence across multiple fungal species—-from the genera *Auricularia, Ganoderma, Grifola, Agaricus, Hericium, Pleurotus, Cordyceps, Ophiocordyceps, Lentinula, Inonotus*, and *Monascus*—that extracts could promote longevity in model organisms. Nonetheless, even though these fungi provide a very diverse source of molecules, that exert activities ranging from disease treatment to life-prolongation, the field of studying how fungal extracts influence longevity is still in its infancy. Therefore, some crucial aspects must be taken into consideration before moving on to definitive conclusions in this review, which may be prioritized in future studies. These include: (1) standardization of extracts should be prioritized. Slight variation in differing lab extraction protocols, or fungal growth protocols, can lead to variation in the contents of extracts. This inherently makes reproducibility and comparison between studies a challenge, which should be resolved to mature the field. (2) Identification of isolated bioactive compounds within extracts should also be prioritized. While a 'single-molecule' medicine may go against the philosophy of multi-component extracts that are the foundation of traditional medicines, it nonetheless allows evaluation in a reproducible manner for the effects of such molecules and deserves a parallel line of inquiry. Finally, (3) Dose makes the medicine. While initial studies in model organisms may look fruitful for the effects of fungal extracts on longevity, the doses used may prove to be untranslatable to humans or produce undesirable toxic side effects. Indeed, even in model organisms, certain studies in this review possessed doses of fungal extracts that either increased lifespan, did nothing, or worse, decreased lifespan. This is consistent with a hormesis response, where low doses of a fungal extract (or another stress) can result in health benefits by activating the activity of stress-resistance genes, producing a net benefit to the organism, while higher doses may overwhelm the organism's stress-resistance mechanisms and become toxic. Therefore, dosing is of paramount importance. Furthermore, these dosing schemes should also take sex into account, as

differences in lifespan effects in some of the studies mentioned in this review were highly variable dependent on the sex involved. We believe that standardization of fungal growing/extraction protocols, identification of sub-fractions or single molecules within extracts, and development of dosing schemes that take sex into account will accelerate the field substantially.

Another important consideration is that the studies presented in this review are from model organisms that are generally short lived, chosen for laboratory research to allow for lifespan studies. These species, not optimized for longevity, may be more responsive to interventions—such as fungi—that activate stress responses and produce hormetic effects. It may therefore be questioned whether fungi could bring the same benefits to humans, an already long-lived organism. Fortunately, epidemiological studies do indeed suggest several roles fungi can play in healthy aging in humans, preserving the quality of physical and mental health in life. One study conducted in Singapore assessing 663 older individuals found that individuals eating > 2 portions per week of mushrooms had reduced chances of having mild cognitive impairment (Feng et al., 2019). Another study, also from Singapore, evaluated how levels of ergothioneine (ET) changed with age, and if a link existed between lower levels of ET in human blood plasma and age-related deterioration of neural functioning (Cheah et al., 2016). Researchers found several interesting observations: blood ET levels decreased with age, and the correlation between mild cognitive impairment and lower blood ET level was significant, implying that ET plays an important role in neuroprotection. In this case fungi can serve as one of the most prominent sources of ET for humans, being one of the few organisms which are able to synthesise it. Additionally, several studies have suggested neuronal health-promoting properties of *H. erinaceus* in individuals affected with mood disorders, Alzheimer's disease and/or accompanying cognitive impairment (Li et al., 2020; Vigna et al., 2019). Although very large-scale human trials are yet to be conducted, the amount of background research is gradually increasing, and are suggestive that humans too can receive health benefit from fungi.

This review holds several limitations. Firstly, studies were included in this review stringently, based either on their inclusion of lifespan assays in natural aging models or their identification of the reversal of aging phenotypes in naturally aged animals. Our review, therefore, did not include articles that studied a fungal compound or extracts that evaluated solely an age-related disease model, an artificially induced aging model, or indirectly studied aging by evaluating, for example, antioxidant effects, and extrapolating this to 'anti-aging' effects. Therefore, other fungal species and extracts may certainly exist that will influence aging, though have not been described in this review. Secondly, many fungal species and extracts included in this review are based on a limited number of studies, sometimes one, and sometimes simple in their design. Therefore, caution should be taken into account with the temptation to extrapolate these findings to human relevance.

To conclude, there is much promising evidence that fungi can serve as a source of bioactive molecules for the development of longevity medicines. There are clear lines of evidence across diverse phylum of fungi, described in various aging models ranging from worms to flies to mice, that bioactive compounds exist in these extracts that produce beneficial effects on longevity. There is a need for reproducing initial findings and a gap exists in evaluating effects on the human aging process. Therefore, the current state of knowledge serves to point researchers in the direction of future studies but has not revealed a singular definitive conclusion about lifespan and healthspan prolonging effects for humans. Nonetheless, the field is young and the stakes are high. We intend for this review to illustrate the promising aspects of fungal extracts as sources of longevity medicines and to guide future studies geared towards translating findings in model organisms to human trials.

Declaration of Competing Interest

None.

Data Availability

No data was used for the research described in the article.

Acknowledgements

GEJ is supported by a VENI grant from ZonMw, and AGEM Talent and Development Grants.

References

Agrawal, S., Samanta, S., Deshmukh, S.K., 2022. The antidiabetic potential of endophytic fungi: future prospects as therapeutic agents. Biotechnol. Appl. Biochem. 69.
Aranaz, P., Peña, A., Vettorazzi, A., Fabra, M.J., Martínez-Abad, A., López-Rubio, A., Pera, J., Parladé, J., Castellari, M., Milagro, F.I., González-Navarro, C.J., 2021. Grifola frondosa (Maitake) extract reduces fat accumulation and improves health span in C. elegans through the daf-16/foxo and skn-1/nrf2 signalling pathways. Nutrients 13.
Bailly, C., Gao, J.M., 2020. Erinacine A and related cyathane diterpenoids: molecular diversity and mechanisms underlying their neuroprotection and anticancer activities. Pharm. Res. 159.
Boddy, L., Crockatt, M.E., Ainsworth, A.M., 2011. Ecology of Hericium cirrhatum, H. coralloides and H. erinaceus in the UK. Fungal Ecol. 4.
Chang, C.J., Lin, C.S., Lu, C.C., Martel, J., Ko, Y.F., Ojcius, D.M., Tseng, S.F., Wu, T.R., Chen, Y.Y.M., Young, J.D., Lai, H.C., 2015. Ganoderma lucidum reduces obesity in mice by modulating the composition of the gut microbiota. Nat. Commun. 6.
Chaudhary, R., Kalra, R., Goel, M., 2020. Role of fungi in neurodegenerative diseases. In: New and Future Developments in Microbial Biotechnology and Bioengineering.
Cheah, I.K., Halliwell, B., 2012. Ergothioneine; antioxidant potential, physiological function and role in disease. Biochim. Biophys. Acta - Mol. Basis Dis. 1822.
Cheah, I.K., Feng, L., Tang, R.M.Y., Lim, K.H.C., Halliwell, B., 2016. Ergothioneine levels in an elderly population decrease with age and incidence of cognitive decline; a risk factor for neurodegeneration? Biochem. Biophys. Res. Commun. 478.
Chen, H., Ju, Y., Li, J., Yu, M., 2012. Antioxidant activities of polysaccharides from Lentinus edodes and their significance for disease prevention. Int. J. Biol. Macromol. 50.
Chen, S., Liu, C., Huang, X., Hu, L., Huang, Y., Chen, H., Fang, Q., Dong, N., Li, M., Tang, W., Nie, S., 2020. Comparison of immunomodulatory effects of three polysaccharide fractions from Lentinula edodes water extracts. J. Funct. Foods 66.
Chen, S.Y., Ho, K.J., Hsieh, Y.J., Wang, L.T., Mau, J.L., 2012. Contents of lovastatin, γ-aminobutyric acid and ergothioneine in mushroom fruiting bodies and mycelia. LWT - Food Sci. Technol. 47.
Chen, Y., Huang, Y., Cui, Z., Liu, J., 2015. Purification, characterization and biological activity of a novel polysaccharide from Inonotus obliquus. Int. J. Biol. Macromol. 79.
Chuang, M.H., Chiou, S.H., Huang, C.H., Yang, W., Bin, Wong, C.H., 2009. The lifespan-promoting effect of acetic acid and Reishi polysaccharide. Bioorg. Med. Chem. 17.
Cuong, V.T., Chen, W., Shi, J., Zhang, M., Yang, H., Wang, N., Yang, S., Li, J., Yang, P., Fei, J., 2019. The anti-oxidation and anti-aging effects of Ganoderma lucidum in Caenorhabditis elegans. Exp. Gerontol. 117.
Fang, J., Wang, Y., Lv, X., Shen, X., Ni, X., Ding, K., 2012. Structure of a β-glucan from Grifola frondosa and its antitumor effect by activating Dectin-1/Syk/NF-κB signaling. Glycoconj. J. 29.
Fang, Z., Chen, Y., Wang, G., Feng, T., Shen, M., Xiao, B., Gu, J., Wang, W., Li, J., Zhang, Y., 2019. Evaluation of the antioxidant effects of acid hydrolysates from: Auricularia auricular polysaccharides using a Caenorhabditis elegans model. Food Funct. 10.
Feng, L., Cheah, I.K.M., Ng, M.M.X., Li, J., Chan, S.M., Lim, S.L., Mahendran, R., Kua, E. H., Halliwell, B., 2019. The association between mushroom consumption and mild cognitive impairment: a community-based cross-sectional study in Singapore. J. Alzheimer's Dis. 68.
Finimundy, T.C., Dillon, A.J.P., Henriques, J.A.P., Ely, M.R., 2014. A review on general nutritional compounds and pharmacological properties of the *Lentinula edodes* mushroom. Food Nutr. Sci. 05.
Flurkey, K., Currer, J.M., Harrison, D.E., 2007. Mouse models in aging research. In The Mouse in Biomedical Research.
Gems, D., Partridge, L., 2013. Genetics of longevity in model organisms: debates and paradigm shifts. Annu. Rev. Physiol. 75.
Gründemann, C., Reinhardt, J.K., Lindequist, U., 2020. European medicinal mushrooms: do they have potential for modern medicine? – an update. Phytomedicine 66.
Guo, L., Xie, J., Ruan, Y., Zhou, L., Zhu, H., Yun, X., Jiang, Y., Lü, L., Chen, K., Min, Z., Wen, Y., Gu, J., 2009. Characterization and immunostimulatory activity of a polysaccharide from the spores of Ganoderma lucidum. Int. Immunopharmacol. 9.
Huang, M., Liu, J., Zhang, S., Mei, X., Yang, X., 2011. Effects of bioactive extracts from four edible mushrooms on the lifespan of Drosophila melanogaster. Mycology 2.
Krakowska, A., Zięba, P., Włodarczyk, A., Kała, K., Sułkowska-Ziaja, K., Bernaś, E., Sękara, A., Ostachowicz, B., Muszyńska, B., 2020. Selected edible medicinal mushrooms from Pleurotus genus as an answer for human civilization diseases. Food Chem. 327.

Kurtzman, R.H., 2005. Mushrooms: sources for modern western medicine. Micol. Apl. Int. 17.
Lai, J.R., Hsu, Y.W., Pan, T.M., Lee, C.L., 2021. Monascin and ankaflavin of monascus purpureus prevent alcoholic liver disease through regulating ampk-mediated lipid metabolism and enhancing both anti-inflammatory and anti-oxidative systems. Molecules 26.
Lee, C.L., Kung, Y.H., Wu, C.L., Hsu, Y.W., Pan, T.M., 2010. Monascin and ankaflavin act as novel hypolipidemic and high-density lipoprotein cholesterol-raising agents in red mold dioscorea. J. Agric. Food Chem. 58.
Lemieszek, M., Rzeski, W., 2012. Anticancer properties of polysaccharides isolated from fungi of the Basidiomycetes class. Wspolczesna Onkol. 16.
Li, I.C., Chen, Y.L., Lee, L.Y., Chen, W.P., Tsai, Y.T., Chen, C.C., Chen, C.S., 2014. Evaluation of the toxicological safety of erinacine A-enriched Hericium erinaceus in a 28-day oral feeding study in Sprague-Dawley rats. Food Chem. Toxicol. 70.
Li, I.C., Lee, L.Y., Chen, Y.J., Chou, M.Y., Wang, M.F., Chen, W.P., Chen, Y.P., Chen, C.C., 2019. Erinacine A-enriched Hericium erinaceus mycelia promotes longevity in Drosophila melanogaster and aged mice. PLoS One 14.
Li, I.C., Chang, H.H., Lin, C.H., Chen, W.P., Lu, T.H., Lee, L.Y., Chen, Y.W., Chen, Y.P., Chen, C.C., Lin, D.P.C., 2020. Prevention of early Alzheimer's disease by erinacine A-enriched hericium erinaceus mycelia pilot double-blind Placebo-controlled study. Front. Aging Neurosci. 12.
Liang, B., Guo, Z., Xie, F., Zhao, A., 2013. Antihyperglycemic and antihyperlipidemic activities of aqueous extract of Hericium erinaceus in experimental diabetic rats. BMC Complement. Altern. Med. 13.
Lin, Q., Hou, S., Dai, Y., Jiang, N., Lin, Y., 2020. Monascin exhibits neuroprotective effects in rotenone model of Parkinson's disease via antioxidation and anti-neuroinflammation. Neuroreport.
Linford, N.J., Bilgir, C., Ro, J., Pletcher, S.D., 2013. Measurement of lifespan in Drosophila melanogaster. J. Vis. Exp.
Lithgow, G.J., Walker, G.A., 2002. Stress resistance as a determinate of C. elegans lifespan. Mech. Ageing Dev. 123.
Liu, P., Xue, J., Tong, S., Dong, W., Wu, P., 2018. Structure characterization and hypoglycaemic activities of two polysaccharides from Inonotus obliquus. Molecules 23.
Liu, X., Huang, Y., Chen, Y., Cao, Y., 2016. Partial structural characterization, as well as immunomodulatory and anti-aging activities of CP2-c2-s2 polysaccharide from: Cordyceps militaris. RSC Adv. 6.
López-Otín, C., Blasco, M.A., Partridge, L., Serrano, M., Kroemer, G., 2013. The hallmarks of aging. Cell 153.
Maity, G.N., Maity, P., Khatua, S., Acharya, K., Dalai, S., Mondal, S., 2021. Structural features and antioxidant activity of a new galactoglucan from edible mushroom Pleurotus djamor. Int. J. Biol. Macromol. 168.
Martel, J., Chang, S.H., Wu, C.Y., Peng, H.H., Hwang, T.L., Ko, Y.F., Young, J.D., Ojcius, D.M., 2021. Recent advances in the field of caloric restriction mimetics and anti-aging molecules. Ageing Res. Rev. 66.
Matjuskova, N., Azena, E., Serstnova, K., Muiznieks, I., 2014. The influence of the hot water extract from shiitake medicinal mushroom, Lentinus edodes (Higher Basidiomycetes) on the food intake, life span, and age-related locomotor activity of Drosophila melanogaster. Int. J. Med. Mushrooms 16.
Moradali, M.F., Mostafavi, H., Ghods, S., Hedjaroude, G.A., 2007. Immunomodulating and anticancer agents in the realm of macromycetes fungi (macrofungi). Int. Immunopharmacol. 7.
Mori, K., Inatomi, S., Ouchi, K., Azumi, Y., Tuchida, T., 2009. Improving effects of the mushroom Yamabushitake (Hericium erinaceus) on mild cognitive impairment: a double-blind placebo-controlled clinical trial. Phyther. Res. 23.
Nagata, A., Tajima, T., Uchida, M., 2006. Supplemental anti-fatigue effects of Cordyceps sinensis (Tochu-Kaso) extract powder during three stepwise exercise of human. Jpn. J. Phys. Fit. Sports Med.
Nguyen, T.L., Chen, J., Hu, Y., Wang, D., Fan, Y., Wang, J., Abula, S., Zhang, J., Qin, T., Chen, X., Chen, X., Khakame, S.K., Dang, B.K., 2012. In vitro antiviral activity of sulfated Auricularia auricula polysaccharides. Carbohydr. Polym. 90.
Nguyen, T.L., Wang, D., Hu, Y., Fan, Y., Wang, J., Abula, S., Guo, L., Zhang, J., Khakame, S.K., Dang, B.K., 2012. Immuno-enhancing activity of sulfated Auricularia auricula polysaccharides. Carbohydr. Polym. 89.
Niwa, Y., Matsuura, H., Murakami, M., Sato, J., Hirai, K., Sumi, H., 2013. Evidence that naturopathic therapy including Cordyceps sinensis prolongs survival of patients with hepatocellular carcinoma. Integr. Cancer Ther. 12.
Pan, H.Y., Ye, Z.W., Zheng, Q.W., Yun, F., Tu, M.Z., Hong, W.G., Chen, B.X., Guo, L.Q., Lin, J.F., 2022. Ergothioneine exhibits longevity-extension effect in Drosophila melanogaster via regulation of cholinergic neurotransmission, tyrosine metabolism, and fatty acid oxidation. Food Funct. 13.
Peng, H.H., Wu, C.Y., Hsiao, Y.C., Martel, J., Ke, P.Y., Chiu, C.Y., Liau, J.C., Chang, I.T.E., Su, Y.H., Ko, Y.F., Young, J.D., Ojcius, D.M., 2021. Ganoderma lucidum stimulates autophagy-dependent longevity pathways in Caenorhabditis elegans and human cells. Aging 13.
Ponnusamy, C., Uddandrao, V.V.S., Pudhupalayam, S.P., Singaravel, S., Periyasamy, T., Ponnusamy, P., Prabhu, P., Sasikumar, V., Ganapathy, S., 2022. Lentinula edodes (edible mushroom) as a nutraceutical: a review. Biosci. Biotechnol. Res. Asia 19, 1–11.
Ratto, D., Corana, F., Mannucci, B., Priori, E.C., Cobelli, F., Roda, E., Ferrari, B., Occhinegro, A., Di Iorio, C., De Luca, F., Cesaroni, V., Girometta, C., Bottone, M.G., Savino, E., Kawagishi, H., Rossi, P., 2019. Hericium erinaceus improves recognition memory and induces hippocampal and cerebellar neurogenesis in frail mice during aging. Nutrients 11.
Roda, E., Ratto, D., De Luca, F., Desiderio, A., Ramieri, M., Goppa, L., Savino, E., Bottone, M.G., Locatelli, C.A., Rossi, P., 2022. Searching for a longevity food, we bump into Hericium erinaceus primordium rich in ergothioneine: the "longevity vitamin" improves locomotor performances during aging. Nutrients 14.
Rosada, A., Kassner, U., Weidemann, F., König, M., Buchmann, N., Steinhagen-Thiessen, E., Spira, D., 2020. Hyperlipidemias in elderly patients: results from the Berlin Aging Study II (BASEII), a cross-sectional study. Lipids Health Dis. 19.
Saitsu, Y., Nishide, A., Kikushima, K., Shimizu, K., Ohnuki, K., 2019. Improvement of cognitive functions by oral intake of hericium erinaceus. Biomed. Res. 40.
Sánchez, J.E., Jiménez-Pérez, G., Liedo, P., 2015. Can consumption of antioxidant rich mushrooms extend longevity?: antioxidant activity of Pleurotus spp. and its effects on Mexican fruit flies' (Anastrepha ludens) longevity. Age 37.
Sasidharan, S., Aravindran, S., Latha, L.Y., Vijenthi, R., Saravanan, D., Amutha, S., 2010. In vitro antioxidant activity and hepatoprotective effects of lentinula edodes against paracetamol-induced hepatotoxicity. Molecules 15.
Scerbak, C., Vayndorf, E.M., Hernandez, A., McGill, C., Taylor, B.E., 2016. Mechanosensory neuron aging: differential trajectories with lifespan-extending alaskan berry and fungal treatments in Caenorhabditis elegans. Front. Aging Neurosci. 8.
Seo, Y.R., Patel, D.K., Shin, W.C., Sim, W.S., Lee, O.H., Lim, K.T., 2019. Structural elucidation and immune-enhancing effects of novel polysaccharide from Grifola frondosa. Biomed. Res. Int. 2019.
Shi, Y.C., Pan, T.M., Liao, V.H.C., 2016. Monascin from monascus-fermented products reduces oxidative stress and amyloid-β toxicity via DAF-16/FOXO in Caenorhabditis elegans. J. Agric. Food Chem. 64.
Shi, Y.X., Chen, W.S., 2020. Monascin ameliorate inflammation in the lipopolysaccharide-induced BV-2 microglial cells via suppressing the NF-κB/p65 pathway. Iran. J. Basic Med. Sci. 23.
Shih, C.C., Lu, M.K., Cheng, J.J., Danny, L.W., 2005. Antiangiogenic activities of polysaccharides isolated from medicinal fungi. FEMS Microbiol. Lett. 249.
Shimbo, M., Kawagishi, H., Yokogoshi, H., 2005. Erinacine A increases catecholamine and nerve growth factor content in the central nervous system of rats. Nutr. Res. 25.
Sillapachaiyaporn, C., Rangsinth, P., Nilkhet, S., Ung, A.T., Chuchawankul, S., Tencomnao, T., 2021. Neuroprotective effects against glutamate-induced ht-22 hippocampal cell damage and Caenorhabditis elegans lifespan/healthspan enhancing activity of auricularia polytricha mushroom extracts. Pharmaceuticals 14.
Sim, Y.C., Lee, J.S., Lee, S., Son, Y.K., Park, J.E., Song, J.E., Ha, S.J., Hong, E.K., 2016. Effects of polysaccharides isolated from Inonotus obliquus against hydrogen peroxide-induced oxidative damage in RINm5F pancreatic cells. Mol. Med. Rep. 14.
Singh, V.K., Patel, Y., Naraian, R., 2012. Medicinal properties of Pleurotus species (oyster mushroom): a review. World J. Fungal Plant Biol. 3, 1–12.
Soares, A.A., de Souza, C.G.M., Daniel, F.M., Ferrari, G.P., da Costa, S.M.G., Peralta, R. M., 2009. Antioxidant activity and total phenolic content of Agaricus brasiliensis (Agaricus blazei Murril) in two stages of maturity. Food Chem. 112.
Soccol, C.R., Bissoqui, L.Y., Rodrigues, C., Rubel, R., Sella, S.R.B.R., Leifa, F., De Souza Vandenberghe, L.P., Soccol, V.T., 2016. Pharmacological properties of biocompounds from spores of the lingzhi or reishi medicinal mushroom Ganoderma lucidum (Agaricomycetes): a review. Int. J. Med. Mushrooms 18.
Sung, G.H., Hywel-Jones, N.L., Sung, J.M., Luangsa-ard, J.J., Shrestha, B., Spatafora, J. W., 2007. Phylogenetic classification of Cordyceps and the clavicipitaceous fungi. Stud. Mycol. 57.
Sutphin, G.L., Kaeberlein, M., 2009. Measuring Caenorhabditis elegans life span on solid media. J. Vis. Exp.
Szychowski, K.A., Skóra, B., Pomianek, T., Gmiński, J., 2021. Inonotus obliquus – from folk medicine to clinical use. J. Tradit. Complement. Med. 11.
Takaku, T., Kimura, Y., Okuda, H., 2001. Isolation of an antitumor compound from Agaricus blazei Murill and its mechanism of action. J. Nutr. 131.
Venturella, G., Ferraro, V., Cirlincione, F., Gargano, M.L., 2021. Medicinal mushrooms: bioactive compounds, use, and clinical trials. Int. J. Mol. Sci. 22.
Vigna, L., Morelli, F., Agnelli, G.M., Napolitano, F., Ratto, D., Occhinegro, A., Di Iorio, C., Savino, E., Girometta, C., Brandalise, F., Rossi, P., 2019. Hericium erinaceus improves mood and sleep disorders in patients affected by overweight or obesity: could circulating pro-BDNF and BDNF be potential biomarkers? Evid.-Based Complement. Altern. Med. 2019.
Wang, B. Sen, Lee, C.P., Chen, Z.T., Yu, H.M., Duh, P. Der, 2012. Comparison of the hepatoprotective activity between cultured Cordyceps militaris and natural Cordyceps sinensis. J. Funct. Foods 4.
Wang, L., Li, J.Q., Zhang, J., Li, Z.M., Liu, H.G., Wang, Y.Z., 2020. Traditional uses, chemical components and pharmacological activities of the genus: Ganoderma P. Karst.: a review. RSC Adv. 10.
Wasser, S.P., 2011. Current findings, future trends, and unsolved problems in studies of medicinal mushrooms. Appl. Microbiol. Biotechnol. 89.
Weng, Y., Xiang, L., Matsuura, A., Zhang, Y., Huang, Q., Qi, J., 2010. Ganodermasides A and B, two novel anti-aging ergosterols from spores of a medicinal mushroom Ganoderma lucidum on yeast via UTH1 gene. Bioorganic Med. Chem. 18.
Wu, T.F., Shi, W.Y., Chiu, Y.C., Chan, Y.Y., 2021. Investigation of the molecular mechanism underlying the inhibitory activities of ethanol extract of Bombyx mori pupa-incubated Cordyceps militaris fruiting bodies toward allergic rhinitis. Biomed. Pharmacother. 135.
Xiaofei, X., Jiguo, Y., Zhengxiang, N., Xuewu, Z., 2017. ^{1}H NMR-based metabolic investigation of the effect of Lentinula edodes-derived polysaccharides on aged mice. J. Food Biochem. 41.
Xu, S., Zhang, Y., Jiang, K., 2016. Antioxidant activity in vitro and in vivo of the polysaccharides from different varieties of Auricularia auricula. Food Funct. 7.
Xu, X., Yan, H., Tang, J., Chen, J., Zhang, X., 2014. Polysaccharides in Lentinus edodes: isolation, structure, immunomodulating activity and future prospective. Crit. Rev. Food Sci. Nutr. 54.

Yoon, S.J., Yu, M.A., Pyun, Y.R., Hwang, J.K., Chu, D.C., Juneja, L.R., Mourão, P.A.S., 2003. The nontoxic mushroom Auricularia auricula contains a polysaccharide with anticoagulant activity mediated by antithrombin. Thromb. Res. 112.
Yuan, B., Zhao, C., Cheng, C., Huang, D. chun, Cheng, S. jie, Cao, C. jiang, Chen, G. tang, 2019. A peptide-Fe(II) complex from Grifola frondosa protein hydrolysates and its immunomodulatory activity. Food Biosci. 32.
Yuan, Q., Xie, F., Tan, J., Yuan, Y., Mei, H., Zheng, Y., Sheng, R., 2022. Extraction, structure and pharmacological effects of the polysaccharides from Cordyceps sinensis: a review. J. Funct. Foods 89.
Zeisel, S.H., 1999. Regulation of "nutraceuticals". Science 80, 285.
Zeng, M., Qi, L., Guo, Y., Zhu, X., Tang, X., Yong, T., Xie, Y., Wu, Q., Zhang, M., Chen, D., 2021. Long-term administration of triterpenoids from Ganoderma lucidum mitigates age-associated brain physiological decline via regulating sphingolipid metabolism and enhancing autophagy in mice. Front. Aging Neurosci. 13.
Zhang, T., Zhao, W., Xie, B., Liu, H., 2020. Effects of Auricularia auricula and its polysaccharide on diet-induced hyperlipidemia rats by modulating gut microbiota. J. Funct. Foods 72.
Zhang, W., Yang, J., Chen, J., Hou, Y., Han, X., 2005. Immunomodulatory and antitumour effects of an exopolysaccharide fraction from cultivated Cordyceps sinensis (Chinese caterpillar fungus) on tumour-bearing mice. Biotechnol. Appl. Biochem. 42.
Zhang, Y., Zeng, Y., Men, Y., Zhang, J., Liu, H., Sun, Y., 2018. Structural characterization and immunomodulatory activity of exopolysaccharides from submerged culture of Auricularia auricula-judae. Int. J. Biol. Macromol. 115.
Zhang, Y., Cai, H., Tao, Z., Yuan, C., Jiang, Z., Liu, J., Kurihara, H., Xu, W., 2021. Ganoderma lucidum spore oil (GLSO), a novel antioxidant, extends the average life span in Drosophila melanogaster. Food Sci. Hum. Wellness 10.
Zhao, S., Rong, C., Liu, Y., Xu, F., Wang, S., Duan, C., Chen, J., Wu, X., 2015. Extraction of a soluble polysaccharide from Auricularia polytricha and evaluation of its anti-hypercholesterolemic effect in rats. Carbohydr. Polym. 122.
Zou, Y., Liu, Y., Ruan, M., Feng, X., Wang, J., Chu, Z., Zhang, Z., 2015. Cordyceps sinensis oral liquid prolongs the lifespan of the fruit fly, Drosophila melanogaster, by inhibiting oxidative stress. Int. J. Mol. Med. 36.